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2D Metal – Organic Frameworks with Topological Insulating behaviour

A Dissertation submitted to Pondicherry University in partial fulfillment of the requirements for the award of the Degree Master of Science in Chemistry

Submitted By

Hemanth.H

Regn.No: 13380034

Under the Supervision of

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Certificate

This is to certify that the dissertation submitted under the title "**2D Metal – Organic Frameworks with Topological Insulating behaviour**" is a bonafide record of the work carried out by **Mr.Hemanth.H** (Regn.No: 13380034) under my guidance, in partial fulfilment of the requirements for the award of the degree of Master of Science in Chemistry during December-2017 to April-2018, at the Department of Chemistry, Pondicherry University.

Dr. Musiri.M.Balakrishnarajan

Project Supervisor

Dr. K. Tharanikkarasu

Head of the Department

Declaration

I, Hemanth.H, hereby declare that the work titled "**2D Metal – Organic Frameworks with Topological Insulating behaviour**" has not been submitted elsewhere for the award of any degree, and is a bonafide report of the work undertaken by me from December-2017 to April-2018 under the supervision of the Dr. Musiri.M.Balakrishna Rajan, Assistant Professor, Department of Chemistry.

(Hemanth.H)

"I think I can safely say that nobody understands Quantum Mechanics"

- Richard Feynmann

Acknowledgement

I extend my heartfelt gratitude to my guide **Dr. M.M.Balakrishna Rajan**, Assistant Professor, Department of Chemistry, Pondicherry University for mentoring me, and supporting me throughout the period of my project.

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(Hemanth.H)

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1 Abstract

Dirac materials, i.e., materials with Dirac cones in their Fermi surface are hard sought as they give massless fermions that leads to fractional quantum Hall effects, ultra-high career mobility and several other interesting phenomena that has promising applications in information sciences and spintronics. The successful characterization of the 2D material graphene, once thought to be unstable in nature serves as the typical example for a typical Dirac material. The band structure of graphene, when viewed from Huckel perspective, should be considered antiaromatic as two of its frontier bands are partially filled, though these bands are degenerate only at the

Dirac point. We herein report a novel two dimensional nanosheet with carbon and nitrogen atoms that has porphyrin like framework made up from tetrazaindacene motif. The network has CMMM two-dimensional symmetry and exhibits an anisotropic Dirac cone in its band structure, with a divalent metal ion at the centres of the 16 membered tetraaza-annulene ring. The flexibility of hosting of a variety of heavier transition metals at the centre and the possibility of isoelectronic replacement of N-H and N groups of the delocalized rings with heavier analogues points to the possibility of tuning the spin-orbit coupling magnitudes that ultimately lead to topological insulators. This also shows that the strategy for rational construction of the two-dimensional π-networks with rare Dirac cones is to start with anti-aromatic/nonaromatic molecular fragments.

2 Introduction

Topological Insulators are a class of materials possessing a non-trivial topology, with the band structure similar to an insulator, with symmetry protected conducting states on the surface, that are responsible for the metallic conductivity displayed by such materials. Such surface states are protected by Time-reversal symmetry, giving rise to chiral edge states.

Quantum Spin Hall Effect, analogous to Hall Effect, splits the spin-up and spin-down electrons, which are counter propagating, giving rise to spin currents[1,2]. But QSH does not require an external

magnetic field for the manifestation of Hall effect[3], rather the Intrinsic magnetic field provided by the metal, by spin-orbit coupling would be sufficient for opening up the band gap at the Fermi Surface, leading to non-trivial topology[4]. This causes a continuous deformation for the Hamiltonian from being gapless to having a finite energy gap at the Fermi[5].

Graphene, a topologically trivial material with two Dirac cones at the Fermi, arising at two asymmetric k-points X and X'[6]. But, the absence of a finite Spin-Orbit coupling in carbon, renders it impossible to convert Graphene to a Topological Insulator. Thus, the initial candidates in the search of Topological Insulators were those with very high spin-orbit coupling, with the first Topological Insulator being reported in CdTe and HgTe quantum wells[7], followed other binary and ternary compounds of heavier *d*, *p* and *f* block elements, with significant spin-orbit coupling[8]. Examples include Bi_2Se_3[9], Bi_2Te_3[10], Sb_2Te_3[11,12], ternary compounds like $TlBiTe_2$,

$TlBiSe_2$[13–17], $PbBi_2Te_4$[18], Heusler compounds[19–21], Bi bilayer[22–28] and actinide based binary compounds.

Organic Topological Insulators, although sounding counter intuitive at first, offers the advantage of easy fabrication, mechanical flexibility and low cost. Presently, two families of OTI's are investigated, one with hexagonal lattice, and the other with Kagome lattice[29]. Examples of OTIs that are presently investigated are 2D metal-triphenyl ([$M(C_6H_5)_3$], M=Pb, Bi), metal-dicyanoanthracene[30], Au-1,3,5-triethynylbenzene(Au-TEB)[31], $X_3(C_{18}H_{12}N_6)_2$[32] where X is 5d transition metals.

3 Computational Methodology

Molecular calculations were performed using Gaussian 09 package[33], at 3.00D-01 level, which corresponds to normal optimization, by employing B3LYP functional at 6-31+g(d) and LANL2DZ basis sets.

NMR calculations were performed using GIAO method at the same level, using Gaussian 09 package[33].

For Material Simulations without considering Spin Orbit coupling, CASTEP[34] bundled with Materials Studio Package was employed. The simulations were performed at Plane Wave cut-off of 400 eV, with ultrasoft pseudopotentials. Phonon calculations were

performed on the optimised geometries of the 2D sheets to check for the stability of the 2D sheets. Brillouin Path was chosen as Gamma - Y - S - Gamma.

For Material Simulations with Spin Obit Coupling, Quantum Espresso[35,36] package was employed. The simulations were performed at Plane Wave cut-off of 60 Ry, with ultrasoft pseudopotentials. Brillouin Path was chosen as Gamma - Y - S - Gamma.

Crystal Orbital Overlaps were computed using bind[37] and viewkel[38] packages available in YAeHMOP. The Brilloiun path was set as Gamma - Y - S - Gamma.

4 Results and Discussion

4.1 Structural Characterisation of Pyrazaporphyrinoids

Porphyrin is a class of water-soluble, nitrogenous biological pigments consisting of four pyrrole rings connected by methine bridges. They are aromatic with 26π electrons present in the system.

Porphyrin network offers a plethora of possibilities for studying the bonding present in the framework, and how the system behaves to external electronic and structural perturbations. The peculiar electronic structure of the porphyrin network with HOMO and HOMO-1 levels being pseudo-degenerate, along with the degenerate LUMO

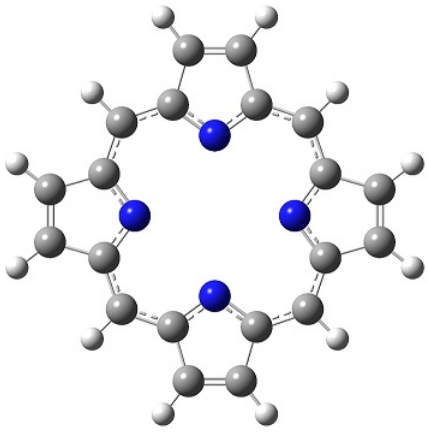

Figure 4.1: *[Porphyrin]$^{2-}$ system*

provides us with an exciting oppurtunity to study the response of the system towards the perturbations applied.

Moving along the same line, we decided to perturb the Porphyrin framework by introducing two pyrazine rings in place of two pyrrole rings, which lowered the symmetry of the system from D_{4h} to D_{2h}, by removing two pairs of C_2 axes from the original system. This structural perturbation ensured that the doubly degenerate LUMO of the [porphyrin]$^{2-}$ system splitting to two pseudo degenrate levels with b_{2g} and a_u symmetry.

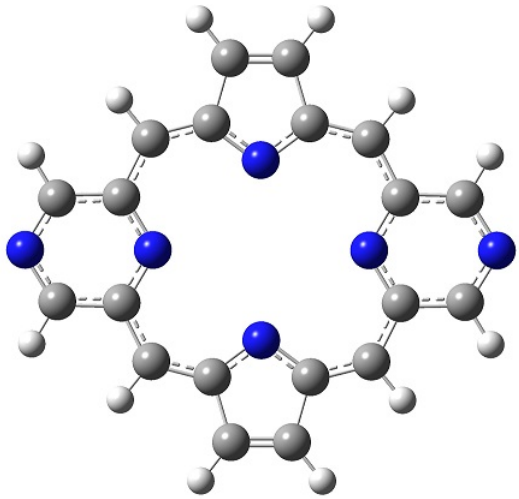

Figure 4.2: *[Pyrazaporphyrin]*$^{2-}$ *system*

An eH analysis showed that the system has it's frontier energy levels lying close in energy, and hence the system can reversibly accept and donate two electrons, while still remaining stable on the potential energy surface. This offered us an exciting possiblity to study about the electronic properties of the system and the extended networks that can be generated from the same.

Calculations showed that the system was stable on the potential energy surface, even after oxidising it by two electrons, or by reduc-

ing it with two electrons. We the decided to look into the structural parameters that have varied as a result of these electronic perturbations.

We found that the central cavity in the case of porphyrinoids were symmetrical, while in the case of pyrazaporphyrinoids, it was found to be elliptical, with the major and minor axes lying along pyrazine and pyrrole rings respectively. This led us to investigate the correlation between the ion size and the cavity size in the pyrazaporphyrinoid structure, which is discussed in further detail in the next section.

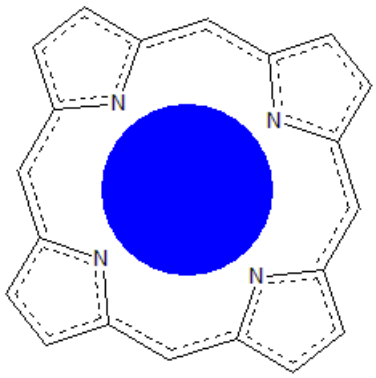

Figure 4.3: *Central Hole in [Porphyrin]*$^{2-}$

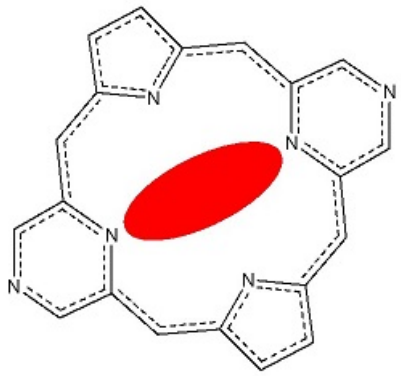

Figure 4.4: *Central Hole in [Porphyrin]$^{2-}$*

4.2 Electronic features of Metallopyrazaporphyrinoids

Different monovalent, divalent and trivalent metals were substituted into the pryazaporphyrinoid framework to study the response of the sytem towards the various electronic perturbations that it could possibly undergo.

[Mg-Pyrazaporphyrin] was found to be a minima on the potential energy surface, with the HOMO and LUMO levels concentrated on

the organic framework, and this property was found to be followed by the heavier analouges also.

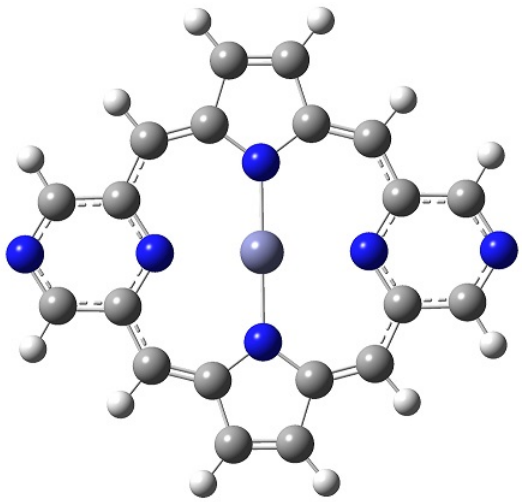

Figure 4.5: *[Zn-Pyrazaporphyrin]*

We found that the pyrazaporphyrin system could host a variety of divalent transition metals in the central hole, without undergoing major geometrical distortions, as manifested by the succesful characterisation of Zn-group and Ni-group pyrazaporphyrins. We then investigated the possibility of having the system reversibly oxidised and reduced by two electrons, and the inferrence that they were able to undergo such electronic changes, without undergoing major dis-

tortions, came as quite surprising. We found that the HOMO and LUMO levels in both Zn-group and Ni-group pyrazaporphyrins had finite contributions from by the d orbitals from the metal center.

The only anomalous result that we found in the transition metal substituted pyrazaporphyrins was that of the Fe-group pyrazaporphyrins being a saddle point on the potential energy surface. It was concluded that the inablity of the ligand system to stabilize the system in d^6 state would have contributed to them being a saddle-point on the potential energy surface. Upon following the highest imaginary frequency, Fe-group pyrazaporphyrins were found to undergo bending, which lowered the symmetry from being D_{2h} to C_{2v}.

[Li-pyrazaporphyrin]$^{1-}$, which is equal to [Mg-pyrazaporphyrin] electronically, was studied to evaluate the effect of ion-size and hole-size in the binding and structural features of pyrazaporphyrin ligand. It was found to be a minima on the potential energy surface.

[Na-pyrazaporphyrin]$^{1-}$ motif, which is electronically equivalent

to [Mg-pyrazaporphyrin], was found to be a first-order saddle point on the potential energy surface. Upon following the imaginary frequency, the Na atom moved out of the central hole in the pyrazaporphyrin ligand, which categorically proved that size of the ion plays an important role in determining the structural stability of the pyrazaporphyrinoid motif.

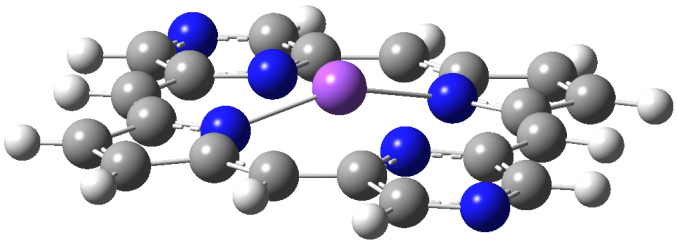

Figure 4.6: *Distorted Na-pyrazaporphyrin*

[Al-pyrazaporphyrin]$^{1-}$, which is equal to [Mg-pyrazaporphyrin]$^{2-}$ electronically, was studied to evaluate the effect of ion-size and hole-size in the binding and structural features of pyrazaporphyrin ligand. It was found to be a second-order saddle point on the potential energy surface. Upon following the highest imaginary frequency, the

pyrazaporphyrin motif underwent bending, to bring the binding sites closer to the metal centre, for effective binding, with the symmetry being lowered to C_{2v}.

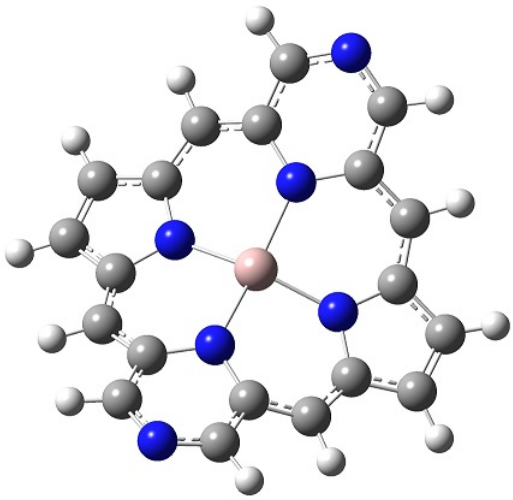

Figure 4.7: *[Al-Pyrazaporphyrin]*$^{1-}$

[Al-pyrazaporphyrin]$^{1+}$, which is equal to [Mg-pyrazaporphyrin] electronically, was studied to evaluate the effect of ion-size and hole-size in the binding and structural features of pyrazaporphyrin ligand. It was found to be a second-order saddle point on the potential energy surface. Upon following the highest imaginary frequency, the pyrazaporphyrin motif underwent ruffling, to bring the binding sites

closer to the metal centre, for effective binding, while the initial symmetry of D_{2h} was lowered to C_{2v}.

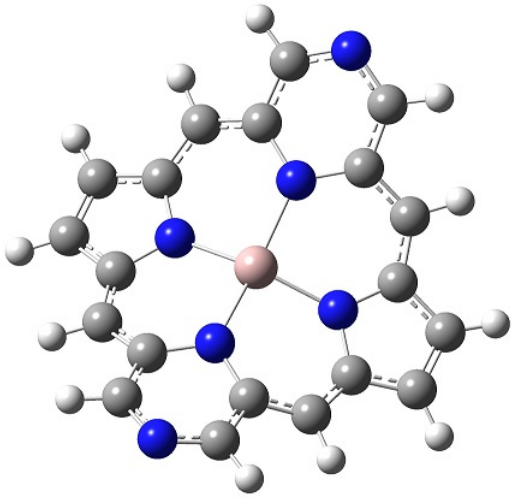

Figure 4.8: *[Al-Pyrazaporphyrin]$^{1+}$*

The distortions in [Al-pyrazaporphyrin]$^{1+}$ and [Al-pyrazaporphyrin]$^{1-}$ were attributed to Pseudo-Jahn Teller Effect, where two or more closely lying orbitals mix, to cause a distortion in the system. The out-of plane distortion was caused by the mixing of HOMO and LUMO+8 energy levels in the [Al-pyrazaporphyrin]$^{1+}$ system.

[Se-pyrazaporphyrin], which is equal to [Mg-pyrazaporphyrin] elec-

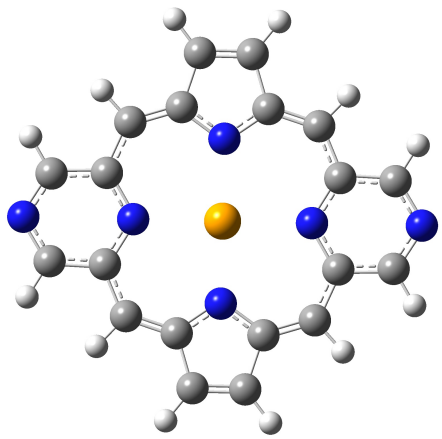

Figure 4.9: *Se-pyrazaporphyrin*

tronically, was studied to evaluate the effect of substituting a non-metal into the pyrazaporphyrin ligand, and how it would manifest itself in dictating the structural and electrical features of the metallopyrazaporphyrin system. It was found to be stable on the potential energy surface, with it being able to withstand two electron oxidation and reduction, which was surprising.

4.3 Extended sheets based on Metallopyrazaporphyrins

Extended sheets based on metallopyazaporphyrins were constructed by connecting the motifs through *meso*-carbon. Different divalent metal ions were substituted into the metallopyrazaporphyrin extended sheets, and the electronic and structural features of the structures were elucidated.

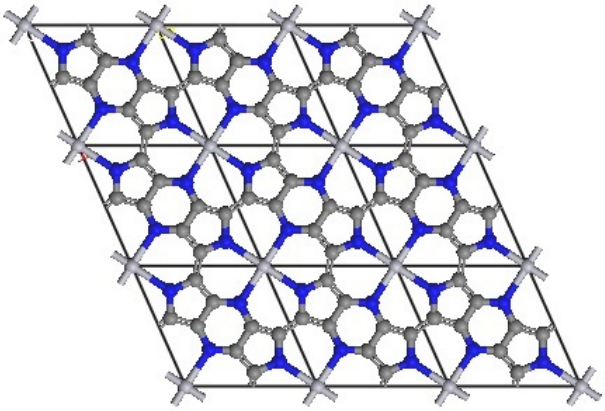

Figure 4.10: *[Hg-pyrazaporphyrin] extended sheet*

Structural Characterisation of the Extended Metallopyrazaporphyrins showed that the major and minor axes were getting inter-

changed upon extending the pyrazaporphyrin motif to a 2D sheet. This interchange was attributed to the *meso*-carbon bridges, which makes the system more rigid in nature. To further investigate the nature of this interchange, the pyrazaporphyrin motif was extended into a molecular fragment, where the *meso*-carbons were tied down by pyrrole rings. We found similar exchange of major and minor axes in the new molecular motif, which confirmed our assumption that *meso-meso* bridges were responsible for the exchange of major and minor axes, due to increased rigidity present in the system.

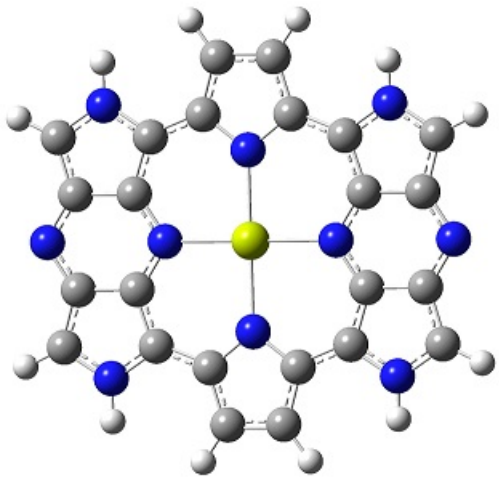

Figure 4.11: *Extended Pyrazaporphyrin with meso-carbons tied down*

It was found that extended sheets based on metallopyrazaporhyrins had an assymetric dirac cone at the Fermi, between Gamma and Y special k-points. The split off bands that gave rise to the dirac cone were mapped to the molecular orbitals from the primitve cell, to give an understanding about the role of the metal in dictating the frontier energy levels of the system.

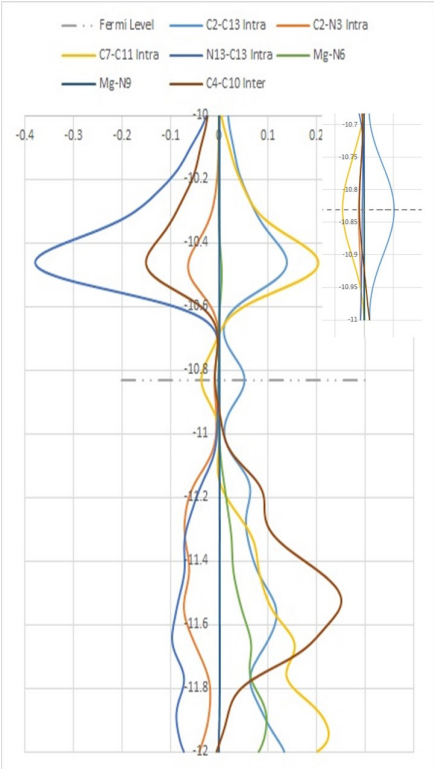

Figure 4.12: *COOP for [Mg-Pyrazaporphyrin] extended system*

A close analysis of the COOP data showed that metal does not play a role in dictating the frontier levels of the extended system, as the contributions form the metal towards the COOPs were found to be very small. It was thus concluded that the Dirac Cone at the Fermi arises purely from the organic framework, with metal acting as a two-electron source for the extended system.

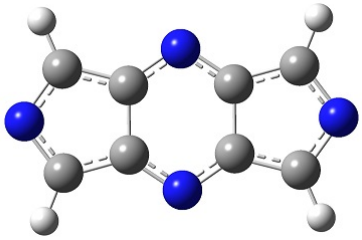

Figure 4.13: *Tetraazaindacene*

The monomer for the extended system was found to be a tetraazaindacene motif, with C_{2h} symmetry, with the system having an open-shell singlet electronic structure, which showed that the monomer had a substantial diradical[39] character, while literature reports suggest that the iso-electronic *s*-indacene, a diradical species; was found

to have a closed-shell electronic structure with D_{2h} symmetry, under constrained conditions, like having bulky substituents on the cyclopentadiene rings[40]. Thus we expected that the extended structure generated out of the monomer will also fall into a space group that corresponds to D_{2h} symmetry, because the system gets tied down at the cyclopentadiene rings.

It then becomes evident that the monomer, tetraaza-indacene is a quasi-delocalised molecule, with mixed aromatic and anti-aromatic character. This quasi-antiaromatic nature of the monomer gives rise to the prescence of the dirac cone at the Fermi in the extended structure.

The presence of two split-off bands, with the band structure otherwise resembling an insulator, ruled out the opto-electronic applications for the present system under investigation. We chose to focus on the topological insulating behaviour of the system, which manifested itself through the presence of a Dirac cone at the Fermi.

The band structures of Mg, Zn, Cd and Hg-pyrazaporphyrin sheets resembles one another, which is quite expected due to the valence orbital being $2s^2$ in Mg and ns^2 in Zn-group. The band structures of the d^8 systems were found to follow a similar trend, with one virtual band running almost parallel to the Fermi, and the Dirac cone in Ni and Pd- pyrazaporphyrin sheets arising from the intersection of the valence band and the band lying immediately above the conduction band. In the case of Pt-pyrazaporphyrin sheet, the virtual band that was running parallel to the Fermi went up in energy, thus the Dirac cone in Pt-pyrazaporphyrin sheet arose from the valence and conduction bands, and not from the higher levels.

To investigate the Topological Insulator behaviour of the pyrazaporphyrin sheets, spin-orbit coupled electronic structure calculations were performed on Hg and Pt-pyrazaporphyrin extended sheets, since they are relativistically active and could alter the band structure at the Fermi, owing to the strong spin-orbit couplling present in them.

It was observed that Spin-Orbit coupling transforms the gapless states at the Fermi in Hg and Pt-pyrazaporphyrin extended sheets, to having a finite energy gap, leading to the opening up of band-gap at the Fermi for such systems.

We found that in Hg-pyrazaporphyrin extended sheet, spin-orbit coupling led the band-gap at the Fermi to open up to 0.038 eV, while Pt-pyrazaporphyrin extended sheet had a band-gap opening of 0.11 eV.

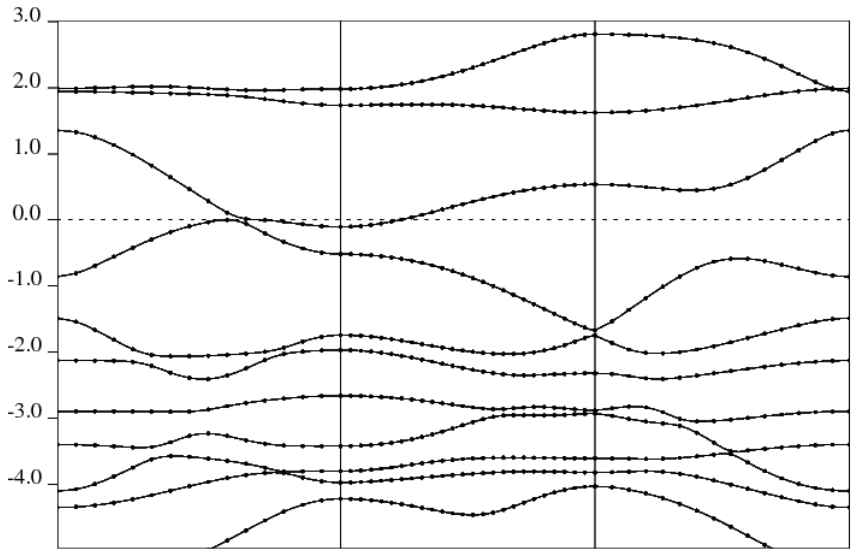

Figure 4.14: *Spin-Orbit coupled Bandstructure for [Pt-Pyrazaporphyrin] extended system*

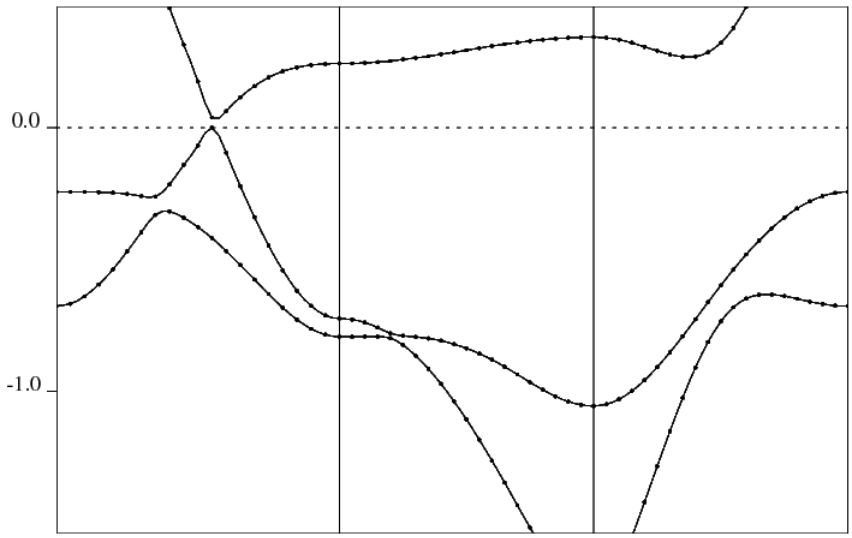

Figure 4.15: *Spin-Orbit coupled Bandstructure for [Hg-Pyrazaporphyrin] extended system*

Phonon calculations on the above optimised 2D sheets showed that [Ni-pyrazaporphyrin], [Mg-pyrazaporphyrin] were stable, while [Cd-pyrazaporphyrin], [Hg-pyrazaporphyrin] and [Pt-pyrazaporphyrin] had a slight out-of plane distortion at the Γ point, and the distorted 2D sheet was found to be approx. 0.10-0.23 kcal/mol lower in energy than the undistorted ones, and the Dirac cone moved slightly above the Fermi in the distorted sheets.

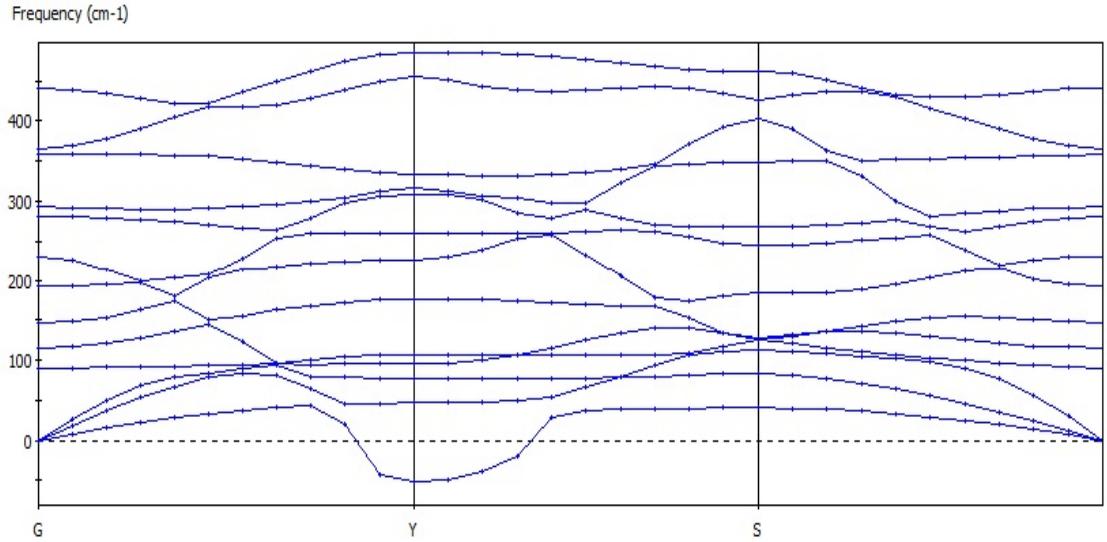

Figure 4.16: *Phonons of [Hg-Pyrazaporphyrin] extended system*

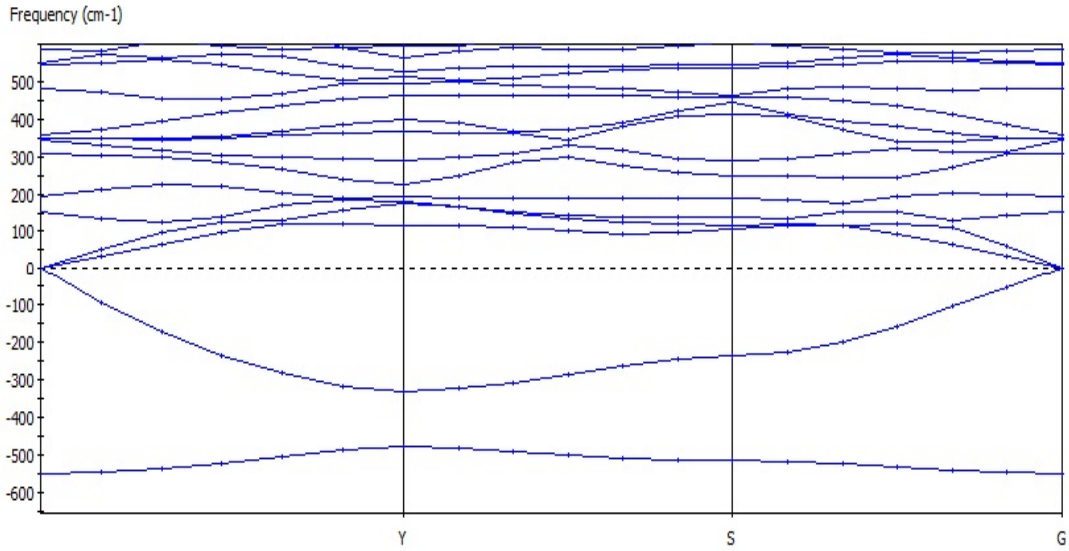

Figure 4.17: *Phonons of [Pt-Pyrazaporphyrin] extended system*

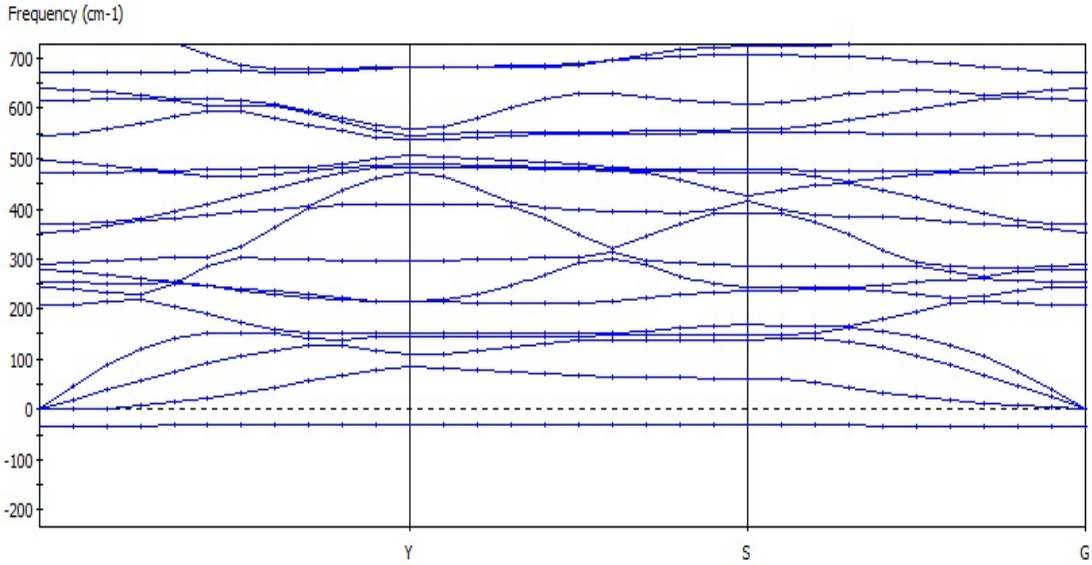

Figure 4.18: *Phonons of [Cd-Pyrazaporphyrin] extended system*

As an ancedote, it must also be noted that the Finite-Displacement method used for calculation of phonon dispersion is not very accurate, and the imaginary frequencies might be an artifact of the inherent errors in the algorith employed for the phonon calculations.

5 Conclusions

A comprehensive study of systems based on pyrazaporphyrin motifs were undertaken. We were able to conclude that extended systems with pyrazaporphyrin motifs could act as Organic Topological Insulators, with suitable central metal atom to provide sufficient spin-orbit coupling to open up the band gap at the Fermi. We could also find an unusual similarity between an ns^2 element and an ns^2np^4 system, as manifested by the similar band structures of [Mg-pyrazaporphyrin] and [Se-pyrazaporphyrin] extended sheets.

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6 Supporting Information

6.1 Sample input file for Quantum Espresso Calculation

The keywords for calculation can be set to 'vc-relax' for full geometry optimisation, while 'scf' does a single-point energy calculation. For spin-orbit coupled calculations, we have to perfom the calculation with noncollinear magnetisation and spin-orbit coupling turned on. These caluclations can be performed as

```
pw.x < input-file.in > output-file.out
```

```
&CONTROL
                     title = 'Hg-Pyrazaporphyrin' ,
                calculation = 'vc-relax' ,
               restart_mode = 'from_scratch' ,
                     outdir = './tmp' ,
                  pseudo_dir = '../../pseudo' ,
                     prefix = 'Hg-Pyrazaporphyrin' ,
                   verbosity = 'high' ,
              etot_conv_thr = 0.001 ,
              forc_conv_thr = 0.01 ,
    tstress = .true. ,
                   tprnfor = .true. ,
/
&SYSTEM
                     ibrav = 0,
                       nat = 13,
                      ntyp = 3,
```

```
                    ecutwfc = 60 ,
                    ecutrho = 600 ,
      noncolin = .true. ,
                   lspinorb = .true. ,
 /
 &ELECTRONS
           electron_maxstep = 200,
                   conv_thr = 1.0D-7 ,
                mixing_mode = 'plain' ,
                mixing_beta = 0.5 ,
                mixing_ndim = 8,
             diagonalization = 'david' ,
              diago_thr_init = 1e-4 ,
 /
&IONS
                ion_dynamics = 'bfgs' ,
 /
```

```
 &CELL
                cell_dynamics = 'bfgs' ,
                  cell_dofree = 'all' ,
 /
CELL_PARAMETERS (bohr)
     0.000000000   -6.763726672   10.091736593
     0.000000000    6.763726672   10.091736593
   -18.897261329    0.000000000    0.000000000
ATOMIC_SPECIES
    C   12.01100  C.pbe-n-rrkjus_psl.0.1.UPF
    N   14.00700  N.pbe-n-rrkjus_psl.0.1.UPF
   Hg  200.59000  Hg.rel-pbe-dn-rrkjus_psl.0.2.2.UPF
ATOMIC_POSITIONS (crystal)
    C     -0.295856812    0.503205271    0.000000000
    C     -0.432516301   -0.114557590    0.000000000
    C      0.295856812   -0.503205271    0.000000000
    C      0.432516301    0.114557590    0.000000000
```

```
    C      0.503205271   -0.295856812   -0.000000000
    C     -0.114557590   -0.432516301   -0.000000000
    C     -0.503205271    0.295856812   -0.000000000
    C      0.114557590    0.432516301   -0.000000000
    N      0.201600298    0.201600298   -0.000000000
    N     -0.201600298   -0.201600298   -0.000000000
    N     -0.299176355    0.299176355   -0.000000000
    N      0.299176355   -0.299176355   -0.000000000
   Hg     -0.000000000    0.000000000   -0.000000000
K_POINTS crystal_b
4
0.000 0.000 0.000 20 !G
0.500 0.500 0.000 20 !Y
0.000 0.500 0.000 20 !S
0.000 0.000 0.000 00 !G
```

6.2 Input file for bandstructure calculation

This caluclation can be performed as

```
bands.x < input-file.in > output-file.out
```

```
&bands
   prefix  = 'Hg-Pyrazaporphyrin'
   filband = 'bands.dat'
   outdir = './tmp/'
/
```

6.3 Input file for plotting bandstructure

This caluclation can be performed as

```
plotband.x < input-file.in > output-file.out
```

```
bands.dat

-1.00 1.00

bands.xmgr

bands.ps

0.5405

1.00 0.5405
```

6.4 Optimised coordinates for Hg-pyrazaporphyrin extended sheet

```
CELL_PARAMETERS (bohr)

     0.000000000   -6.763726672   10.091736593

     0.000000000    6.763726672   10.091736593

   -18.897261329    0.000000000    0.000000000


ATOMIC_POSITIONS (crystal)

C       -0.296744388   0.506144330  -0.000000000
```

C	-0.428867199	-0.113809012	0.000000000
C	0.296744388	-0.506144330	-0.000000000
C	0.428867199	0.113809012	0.000000000
C	0.506144330	-0.296744388	0.000000000
C	-0.113809012	-0.428867199	-0.000000000
C	-0.506144330	0.296744388	0.000000000
C	0.113809012	0.428867199	0.000000000
N	0.197844136	0.197844136	0.000000000
N	-0.197844136	-0.197844136	0.000000000
N	-0.303009198	0.303009198	-0.000000000
N	0.303009198	-0.303009198	0.000000000
Hg	0.000000000	-0.000000000	0.000000000

6.5 Optimised coordinates for Pt-pyrazaporphyrin extended sheet

```
CELL_PARAMETERS (bohr)
   0.000000000  -6.763726678  10.091736602
   0.000000000   6.763726678  10.091736602
 -18.897261346   0.000000000   0.000000000


ATOMIC_POSITIONS (crystal)
C       -0.294892565   0.503583101  -0.000000000
C       -0.426245250  -0.112934857   0.000000000
C        0.294892565  -0.503583101  -0.000000000
C        0.426245250   0.112934857  -0.000000000
C        0.503583101  -0.294892565  -0.000000000
C       -0.112934857  -0.426245250   0.000000000
C       -0.503583101   0.294892565  -0.000000000
C        0.112934857   0.426245250   0.000000000
```

N	0.192340511	0.192340511	0.000000000
N	-0.192340511	-0.192340511	0.000000000
N	-0.290160813	0.290160813	0.000000000
N	0.290160813	-0.290160813	0.000000000
Pt	0.000000000	0.000000000	0.000000000

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